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B. TECH PROJECT SYNOPSIS

Project ID:A03

Project Topic: “WildEcoWatch : An Integrated IoT and AI Framework for Wildlife Surveillance and Conservation”

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ABSTRACT

WildEcoWatch is an integrated IoT and AI framework designed for wildlife surveillance and conservation. The system utilizes motion detection sensors, camera modules, and AI-based object detection to monitor wildlife activity in real-time. When motion is detected, the system captures images, processes them using a pre-trained machine learning model to identify animal species, and logs the data for analysis. This automated approach helps in tracking wildlife movements without human intervention, making it valuable for conservation efforts in protected areas.

The core of WildEcoWatch consists of a Raspberry Pi-based IoT setup with a PIR (Passive Infrared) sensor to detect motion, triggering a camera to capture images. The captured images are analyzed using AI model, which predicts detected animals with confidence scores. If a recognized animal is found, the system activates visual and auditory alerts while logging the detection details in a CSV file for further study.

To enhance real-time monitoring, the system integrates a Flask-based web server that provides a user interface for manual image capture and detection. Additionally, email alerts are sent to conservationists or researchers when significant detections occur, ensuring prompt response to wildlife activity. The framework is designed to operate autonomously in remote locations, making it a cost-effective solution for wildlife monitoring.

By combining IoT sensors, AI-based recognition, and automated logging, WildEcoWatch offers a scalable and efficient approach to wildlife surveillance. It reduces human dependency in data collection, minimizes disturbance to animals, and provides valuable insights for ecological research and habitat protection. This system can be deployed in national parks, sanctuaries, or research sites to aid in biodiversity conservation efforts.

INTRODUCTION

1.1 Background

Wildlife conservation is a critical challenge due to habitat loss, poaching, and human-wildlife conflicts. Traditional monitoring methods rely on manual observation, which is time-consuming, labor-intensive, and often ineffective in remote areas. Advances in IoT (Internet of Things) and AI (Artificial Intelligence) provide an opportunity to automate wildlife surveillance, enabling real-time detection and tracking of animal movements. WildEcoWatch leverages these technologies to create an integrated system that enhances monitoring efficiency while minimizing human intervention.

1.2 Problem Statement

Current wildlife monitoring techniques suffer from limitations such as delayed detection, lack of real-time alerts, and dependency on manual data collection. This leads to gaps in tracking animal behavior, illegal activities, and ecological changes. A scalable, automated solution is needed to improve accuracy, reduce response time, and provide reliable data for conservation efforts. WildEcoWatch addresses these challenges by combining motion sensors, AI-based image recognition, and automated logging to create a robust surveillance system.

1.3 Motivation / Need of project

The increasing threats to wildlife demand innovative conservation strategies. WildEcoWatch aims to bridge the gap between technology and ecology by providing a cost-effective, automated monitoring system. By detecting and classifying animals in real-time, the system helps researchers and forest authorities track species, prevent poaching, and study habitat usage. The project's motivation stems from the need for sustainable, non-intrusive wildlife monitoring that enhances conservation efforts without disrupting natural ecosystems.

1.4 Objective and Scope of the Project

The main objective of WildEcoWatch is to develop an automated wildlife monitoring system using IoT sensors and AI-based image recognition. The system detects animal movements via a PIR sensor, captures images, and classifies species using a pre-trained deep learning model. It logs detection data (species, timestamp, confidence) and triggers real-time alerts with LED, buzzer and email for immediate response.

The scope includes deploying the system in wildlife zones to monitor animal activity, generate reports, and assist in ecological research. Future enhancements may include integrating GPS tracking and expanding the AI model for more species detection.

LITERATURE SURVEY

2.1 Literature Survey

Sr. No	Title of the Paper	Year of Publication	Publisher	Methodology	Conclusion
1	Animal Intrusion Detection in Farming Area using YOLOv5 Approach	IEEE, ICCAS 2022	(1).Normaisharah Mamat, (2).Mohd Fauzi Othman, (3).Fitri Yakub.	Utilized YOLOv5 , a deep learning model, for detecting animal intrusion (elephant, monkey, tiger, wild boar). Dataset: 280 images (70 per class), resized to 416×416 pixels, split into training (80%), validation (10%), and testing (10%). Evaluation metrics: Precision, Recall, and mean Average Precision (mAP).	(1).The YOLOv5-based approach successfully detects animal intrusion in farming areas with a mean average precision (mAP) of 94%, precision of 89%, and recall of 89%. (2).The results demonstrated YOLOv5's superior performance compared to YOLOv4.
2	Animal Intrusion Detection Using Yolo V8	IEEE, ICACCS 2024	(1).Kathir M, (2).Balaji V , (3).Ashwini K	Implemented YOLOv8, an anchor-free, state-of-the-art object detection model, for identifying four animal classes (buffalo, elephant, rhino, zebra). Dataset: 376 images per class, augmented and processed with bounding boxes. YOLOv8 outperformed earlier YOLO versions with features like mosaic data augmentation and efficient convolutional blocks. Metrics: Achieved mAP of 99.4% on training and 95.9% on validation datasets.	(1).YOLOv8 demonstrated high accuracy and efficiency for real-time animal intrusion detection, promoting human-wildlife coexistence. (2)Future enhancements include species behavior analysis, real-time tracking, and community engagement for conflict mitigation.

3	Implementatio n of a Wild Animal Intrusion Detection Model Based on Internet of Things	IEEE 2022	(1).Prabhat Kumar Panda, (2).Cherlopalli Srujan Kumar, (3).Bommu Sai Vivek, (4).Masura Balachandra, (5).Shashi Kant Dargar	The system uses an ESP32 camera, two Arduino Uno boards, and an E-vehicle prototype. The transmitting Arduino is equipped with ultrasonic sensors and an NRF24L01 transmitter to detect animal intrusions and send data. The receiving Arduino, with an NRF24L01 receiver, controls the E-vehicle, which includes a servo motor, DS1307 timer, and ESP32 camera. When an intrusion is detected, the system activates the vehicle, captures images of the intruder, and initiates motion to respond effectively.	(1)In this paper, wild animal intrusion detection system using Internet of Things has been implemented. (2)current model is built using Arduino uno and various sensors. In the future improvements include use of IR cameras and wide angle cameras to increase system versatility.
4	Enhancing Farm Security: Animal Intrusion Detection using YOLO	IEEE Xplore 2023	(1). M. Venkata Krishna Rao. (2). Aakanksha Challawar (3). Begari Adithi. (4). Akhila Kurmandla. (5). Pranathi Bussa	(1).Utilizes YOLOv3 classifier and OpenCV library for animal detection. (2).Captures images or videos of the environment using high-sensitivity sensors. (3).Processes images for feature extraction and object detection. (4).Classifies detected animals into harmful and harmless categories. (5).Implements an alert system that activates alarms for harmful animal	(1).The animal intrusion detection system provides an effective and safe method for monitoring agricultural activities, enhancing crop management, and increasing productivity. (2).It addresses the challenges of false alarms and accuracy in specific scenarios, emphasizing the need for regular updates and maintenance to ensure optimal performance.
5	Image Processing based	IEEE ICEARS 2022	(1)A.Sathesh, (2)K.Vishnu, (3)A.Yuvanesh	(1)Utilizes a surveillance system with	(1)The proposed system effectively

	Protection of Crops from Wild Animals using Intelligent Surveillance		war, (4)V.Vellaisamy (5)K.Gowthami	cameras fixed around farmlands. (2)Implements YOLO (You Only Look Once) algorithm for object detection and classification. Features include: (3)Pre-processing of captured images. (4)Feature extraction and comparison with predefined images. (5)Use of Arduino and GSM module for alerting farmers via alarms and messages. (6)The system aims to detect and alert about animal intrusions without harming the animals.	captures images of intruding animals and accurately classifies them using the YOLO algorithm. (2)It provides a low-cost, energy-efficient solution to protect crops from wild animals, addressing a significant social issue for farmers. (3)The system enhances agricultural productivity by preventing crop destruction and reducing financial losses for farmers.
6	Image Processing based Animal Intrusion Detection System in Agricultural Field using Deep Learning	IEEE 2024	(1). Oviya Gnanasekar. (2). Prapti Dinesh. (3). Srividhya K	(1).Utilizes Raspberry Pi and deep learning techniques, specifically YOLOv7, for real-time animal detection. (2)Captures live video feeds from agricultural fields and analyzes them to detect animal intrusions. (3).Emits specific repellent sounds and sends SMS alerts to farmers upon detection of intrusions.	YOLOv7 model demonstrates high accuracy and speed in detecting animal intrusions, contributing to the sustainability and productivity of agricultural practices. Future enhancements may include the use of drones and more advanced cameras for improved monitoring.
7	Enhancing Animal Intrusion Detection with Raspberry Pi and Fast R-CNN: A Practical Approach for Wildlife Manageme	IEEE 2024	(1).N Kanthimathi (2). R Sheela Daniel (3). R. Santhana Krishnan. (4). J. Samuel Prabakaran (5). S. Gopikumar (6). V. Sakthi Maragatham	(1)Integration of Raspberry Pi with camera module, thermal sensors, and motion sensors. (2)Utilization of Fast R-CNN model for local video processing. (3)Comparison of Fast R-CNN with YOLO and SSD	The proposed animal intrusion detection system ensures real-time, reliable protection of agricultural and wildlife areas, with Fast R-CNN outperforming YOLO and SSD models in detection

	nt			models across three datasets: Caltech Camera Traps, ImageNet, and Open Images. (4)Evaluation based on accuracy, precision, recall, and F1 score.	accuracy and response.
8	Animals Detection System In The Farm Area Using Iot	ICCCI 2023	(1). Ms.K.Chitra (2).Dr.A.Tamilarasi (3).Dr.M.Pyingkodi (4) K. Nanthini	(1)Utilizes Arduino Uno to control ultrasonic sensors for detecting animal intrusion. (2)The system sends signals to a GSM module to alert farmers via calls and alarms when animals are detected. (3)The ultrasonic sensors are programmed to detect animals at specified heights and trigger alarms. (4)Continuous monitoring allows for alerts until the farmer responds.	(1).The system effectively safeguards crops from wild animals, reducing potential losses for farmers. (2).It provides a reliable method for farmers to monitor their fields without constant supervision, allowing them to focus on other agricultural tasks. (3).The design is affordable and can operate 24/7, enhancing crop protection through IoT technology.
9	Animal Detection and Classification in Image & Video Frames using YOLOv5 and YOLOv8	IEEE 2023	(1). Aishwarya D Shetty (2) Soumya Ashwath	(1)Developed a machine learning system for automatic animal detection in surveillance footage using YOLOv5 and YOLOv8. (2)Utilized the COCO dataset for training and testing the models. (3)Implemented a user interface for inputting images or videos, including live input options. (4)Conducted pre-processing of	(1).YOLOv8 achieved higher accuracy (around 85%) compared to YOLOv5, making it more suitable for applications requiring high precision. (2)YOLOv5, while faster and smaller, had lower accuracy, making it more appropriate for real-time applications where speed is crucial. The choice

				images and created a dataset.	between YOLOv5 and YOLOv8 should (depend on specific application requirement
10	Advanced Wild Animal Detection and Alert System using YOLO V5 Model	IEEE 2023	(1)C.S.Madhumathi (2) Mr.V.Naveen (3) Mr.Nallamala Akshay (4) Mr.M.Sanjay Kumar	(1)Utilizes the YOLO V5 object detection algorithm for real-time wild animal detection. (2)Employs cameras to capture video feeds, which are processed to identify wild animals. (3)Implements an alert system that plays sounds to deter detected animals. (4)Incorporates a user-friendly interface for monitoring and alerts. (5)Evaluates system performance using metrics like recall, accuracy, and precision.	(1)The proposed system offers a cost-effective solution for detecting and tracking wild animals, enhancing community safety and environmental well-being. (2).Future work aims to improve accuracy, integrate with other systems, and expand deployment to diverse regions.
11	Wild Animal Detection using Deep Learning	IEEE 2022	(1). Sreedevi K L (2). Dr Anitha Edison	(1)Utilized depth-wise separable convolution layers for model architecture. (2)Implemented zero padding to maintain edge characteristics and control output image size. (3)Employed the IWildCam dataset for testing, which includes images of various wild animals. (4)Evaluated the model using metrics such as Intersection over Union (IOU), precision, recall, F1 score, and confusion matrix.	(1).The proposed algorithm effectively detects and classifies wild animals, achieving an intersection over union value of 0.878 and an accuracy of 99.6%. (2).The results indicate that the model is powerful in detecting and classifying wild animals, with potential for further improvement through the use of multiple sensors.

12	Crops Protection System from Animals using Arduino	IEEE 2021	(1)P.Manikandan (2) A.Thenmozhi (3). G. Ramesh (4). T. Ravi Kiran Naidu. (5). K. Vamsinath Reddy. (6).K.BhanuPradeep Kumar Reddy	(1)Utilizes Arduino, PIR motion sensor, buzzer, LED lights, and GSM module. (2)Detects animal movement using PIR sensor. (3)Activates buzzer and LED lights upon detection. (4)Sends SMS alerts to the farmer within 10 seconds of detection. (5)Designed to protect crops from animal interference without harming animals or humans.	(1).The proposed system effectively alerts farmers to animal intrusions, helping to prevent crop damage and financial losses. (2).It is practical and affordable for farmers, enhancing crop yields and improving their financial condition. Future enhancements may include the use of cellular networks and additional sensors for better monitoring and protection.
13	Real-Time Monitoring of Agricultural Land with Crop Prediction and Animal Intrusion Prevention using Internet of Things and Machine Learning at Edge	IEEE 2020	(1). Nikhil R (2). Prof. Anisha B (3). Dr. Ramakanth Kumar P	(1)Utilizes IoT devices for smart irrigation based on soil moisture, humidity, and temperature. (2)Implements machine learning techniques for crop prediction based on soil parameters. (3)Employs ultrasonic sound for animal intrusion prevention and alerts farmers via SMS and Email notifications. (4)Data collected from sensors is sent to the cloud for analysis and decision-making.	(1).The proposed system is cost-effective, enhances agricultural water management, aids in crop prediction, and prevents wild animal intrusions. (2).It automates irrigation processes based on real-time soil conditions, thereby improving efficiency and reducing resource wastage.
14	Intrusion Detection and Repellent System for Wild	IEEE 2022	(1). Hardiki Deepak Patil (2). Dr. Namrata Farooq Ansari	(1).Utilizes Raspberry Pi for movement detection and video capture. (2)Implements	(1).The proposed system offers an innovative solution to mitigate human-wildlife conflicts

	Animals Using Artificial Intelligence of Things			YOLOv3 for real-time animal detection. (3)Employs PIR sensors to detect motion and trigger the camera. (4.)Activates a buzzer and flashlight to repel detected wild animals without harm.	by providing a safe and automated method to detect and repel wild animals, thereby protecting both human habitats and wildlife.
15	Animal Intrusion Detection and Ranging system using YOLOv4 and LoRa	IEEE 2022	(1). Dr BMeenakshi (2). Aouthithiye Barathwaj SR Y (3). Haariharan N C (4). Krishnakanth L (5). Abishek J	(1)Integrated system using YOLOv4 for animal detection and LoRa for communication. (2)Utilizes Raspberry Pi for image processing and ESP-32 for communication. (3)Employs K-means clustering for predicting animal locations based on past data.	(1).The proposed system effectively addresses man-animal conflict by providing timely alerts to farmers and wildlife authorities. (2).It aims to create a sustainable ecosystem by minimizing property and life loss for both humans and animals. Successful localization of wild animals enhances response strategies to prevent conflicts.

Summary of Literature survey

Recent research highlights the growing adoption of AI and IoT for wildlife surveillance, with YOLO-based models emerging as a dominant approach due to their real-time detection capabilities. Studies such as [1] (YOLOv5) and [2] (YOLOv8) demonstrate the effectiveness of deep learning in animal intrusion detection, particularly for crop protection and biodiversity conservation. Other works like [9] compare YOLOv5 and YOLOv8 for animal classification, while [10] and [11] further validate the use of YOLO models in wild animal detection systems. These AI-driven approaches provide high accuracy in species recognition, forming a key foundation for WildEcoWatch's detection framework.

IoT-based solutions complement AI by enabling real-time monitoring through sensor networks. Research such as [3] and [8] utilize ultrasonic sensors and GSM modules for intrusion alerts, whereas [7] and [14] integrate Raspberry Pi with thermal and motion sensors for wildlife surveillance. Additionally, [12] and [15] explore hybrid systems combining Arduino, LoRa, and PIR sensors for animal deterrence and long-range communication. These studies emphasize the importance of low-power, scalable IoT devices in remote wildlife monitoring—principles that WildEcoWatch incorporates using PIR sensors and automated alert mechanisms.

Beyond detection, several papers focus on response mechanisms, including alarms, repellent systems, and data logging. Works like [4], [5], and [6] employ image processing and deep learning to trigger deterrents such as buzzers, lights, or GSM alerts, similar to WildEcoWatch's LED/buzzer and email notifications. Studies such as [13] and [14] further enhance these systems with edge computing and AIoT for real-time decision-making. Collectively, these advancements underscore the viability of integrating AI classification, IoT sensors, and automated alerts—core components of WildEcoWatch—to create an efficient, scalable solution for wildlife conservation and human-wildlife conflict mitigation.

3. METHODOLOGY

The **WildEcoWatch** system begins with the initialization of its hardware and software components. The Raspberry Pi is configured to interface with a PIR motion sensor, an LED, a buzzer, and a USB webcam through GPIO pins. Simultaneously, a Flask server is launched to manage backend operations and serve the frontend interface. A dedicated background thread continuously monitors the PIR sensor for any motion in the area.

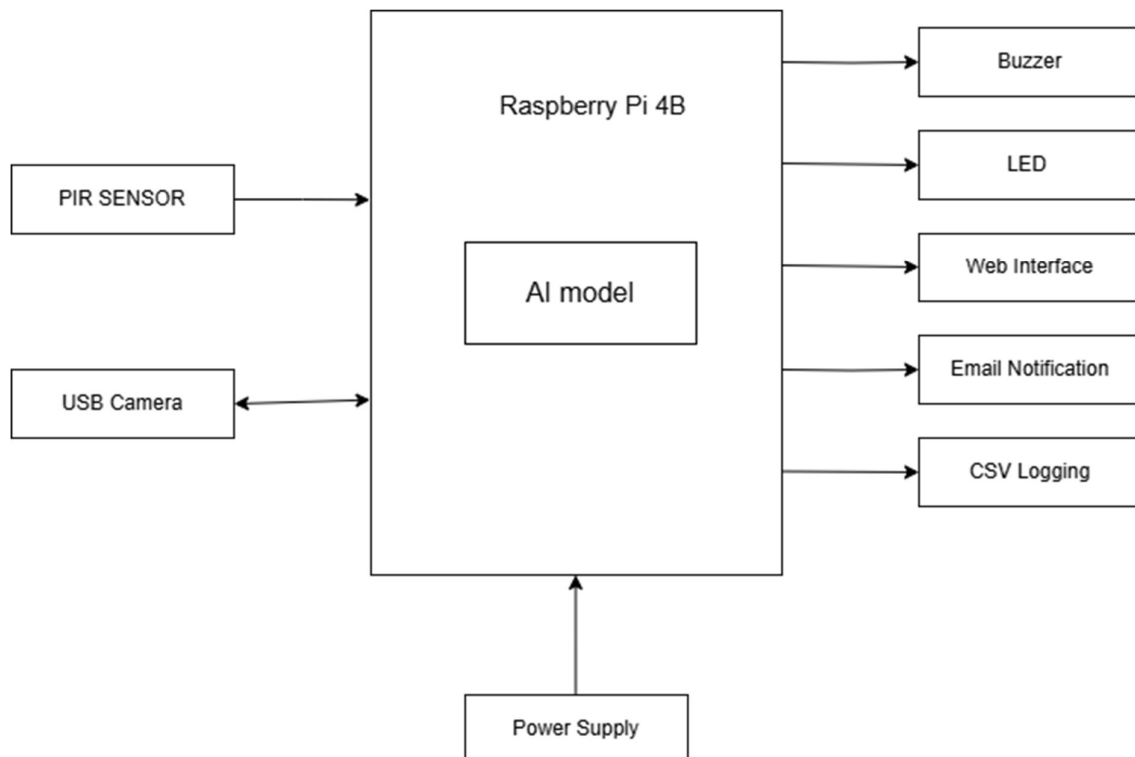
Once the PIR sensor detects motion, the system triggers the webcam to capture a still image of the scene. This image is resized and saved to the local filesystem. To enable compatibility with the Roboflow API, the image is encoded into a Base64 string.

The encoded image is then sent to a custom-trained YOLOv8 model hosted on Roboflow. The model processes the image and returns predictions with animal type and confidence scores. Only those predictions with a confidence level of 70% or higher are considered valid for further actions.

If animals are detected, the system responds by briefly activating the buzzer and LED to alert nearby personnel or deter the animal. An automated email alert containing the detected animal types and their confidence levels is also sent to the recipient using the Mailjet API.

Each detection is logged locally in a CSV file along with the current timestamp, animal type, and confidence percentage. This logging feature provides a reliable record for wildlife monitoring and analysis. Furthermore, users can manually trigger detection and view results through a web interface hosted by the Flask server, which enhances usability and accessibility.

3.1 Block diagram:



The block diagram (Figure 3.1) outlines the major components and operational flow. It can be divided into three major functional blocks:

Unit 1: Sensing and Image Capture

- **PIR Sensor (Motion Detection):**
Continuously monitors the area for any movement using infrared sensing. Once motion is detected, it acts as a trigger to activate the image capture process.
- **USB Camera (Image Acquisition):**
Upon trigger from the PIR sensor, the connected webcam captures an image of the surroundings. This image is forwarded to the processing unit for analysis.

Unit 2: Processing and Intelligence

- **Raspberry Pi 4B (Central Controller):**
Acts as the brain of the system and handles all logic, communication, and decision-making. Its tasks include:
 - i) Capturing images from the camera
 - ii) Sending image data to the AI model hosted via Roboflow for animal detection
 - iii) Receiving predictions and analyzing confidence levels
 - iv) Controlling outputs (LED, buzzer) and sending alerts if animals are detected
 - v) Logging detection data into a CSV file for future reference
 - vi) Hosting a local web interface for user interaction
- **AI Model (YOLOv8 via Roboflow API):**
Processes the captured image to identify wild animals with a high degree of accuracy. Only detections above a certain confidence threshold (e.g., 70%) are considered valid.

Unit 3: Output & Alert Mechanisms

- **LED Indicator:**
Visually signals detection by flashing when an animal is positively identified.
- **Buzzer:**
Sounds an audible alarm to alert nearby individuals when an animal is detected.
- **Email Notification Module (via Mailjet API):**
Automatically sends an email alert with animal type, confidence score, and timestamp to a predefined user email address.
- **CSV Logging:**
Detection details such as date, time, animal type, and confidence are logged locally for analysis, research, or record-keeping.
- **Web Interface:**
A browser-accessible interface allows users to manually trigger detections and view the latest results/images.

Central Support: Power Supply

All components are powered by a stable 5V/3A power source connected to the Raspberry Pi, ensuring uninterrupted system operation in field conditions.

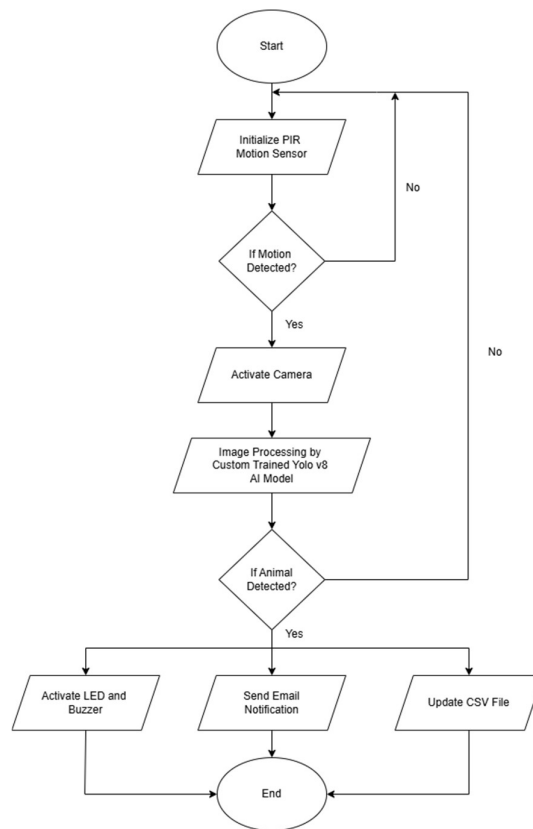
3.2 Hardware specifications / requirements

Component	Specification / Role
Raspberry Pi 4	Central controller that runs the code and connects to sensors
PIR Motion Sensor	Detects motion in the monitored area
USB Webcam	Captures images upon motion detection
LED	Acts as a visual alert when an animal is detected
Buzzer	Provides audible alert upon detection
Jumper Wires	Used to connect components to GPIO pins
Breadboard	For prototyping and organizing sensor connections
Power Supply	To power the Raspberry Pi and connected components
MicroSD Card	Stores the OS and codebase

3.3 Software specifications / requirements

Component	Specification / Role
Raspberry Pi OS	Operating system running on Raspberry Pi
Python 3	Core programming language used for scripting and logic
Flask	Backend web framework to handle API and serve frontend
OpenCV (cv2)	For capturing and processing images from the webcam
Requests	To make HTTP requests to Roboflow API
Mailjet REST API	To send email alerts upon animal detection
Pandas	For handling and logging data in CSV format
Base64	For encoding images before sending to Roboflow API
Threading	For running motion detection in the background
RPi.GPIO	To interface with hardware components via GPIO pins
HTML/CSS/JavaScript	For the frontend user interface served by Flask

3.4 Flow chart



1. Start

The system begins by initializing all hardware and software components, ensuring they are ready for operation.

2. Initialize PIR Motion Sensor

The Passive Infrared (PIR) sensor is activated to detect motion within its range. This sensor is the primary trigger for the system.

3. If Motion Detected?

- The system checks if the PIR sensor has detected any motion.
 - **Yes:** Proceeds to the next step (Activate Camera).
 - **No:** The system remains idle, continuously monitoring for motion.

4. Activate Camera

Upon motion detection, the camera module is triggered to capture an image of the scene where motion was detected.

5. Image Processing by Custom Trained YOLOv8 AI Model

The captured image is processed using a pre-trained YOLOv8 model, which is optimized for wildlife detection. The model analyzes the image to identify and classify any animals present.

6. If Animal Detected?

- The system checks if the AI model has identified an animal in the image.
 - **Yes:** Proceeds to activate alerts and log data.
 - **No:** The system returns to the monitoring state.

7. Activate LED and Buzzer

If an animal is detected, the system triggers visual (LED) and auditory (buzzer) alerts to notify nearby personnel or deter the animal if necessary.

8. Send Email Notification

An email alert is sent to predefined recipients (e.g., researchers or conservationists) containing details of the detected animal, such as species and confidence score.

9. Update CSV File

The detection details (timestamp, animal species, confidence score) are logged into a CSV file for future analysis and reporting.

10. End

The process concludes, and the system resets to continue monitoring for new motion event.

4. CONTRIBUTION OF THE PROJECT

Special Interest Group (SIG): Automation and Communication & Networking

The WildEcoWatch project leverages AI-driven wildlife detection and monitoring to create an advanced system for managing human-wildlife conflict and protecting crops, livestock, and ecological systems.

The project's contributions can be categorized into the following application areas:

1. Health & Hygiene:
By preventing wildlife intrusions near human habitats, the system reduces risks associated with the spread of zoonotic diseases and minimizes physical encounters that could cause injury.
2. Agriculture: Crop, Food, and Water:
The system safeguards crops and water sources from damage caused by wild animals. Farmers can protect their livelihoods by receiving timely alerts and activating deterrent mechanisms, which reduces crop losses and improves agricultural productivity.
3. Energy:
Through the efficient use of solar-powered sensors and energy-efficient AI models, the project supports sustainable energy practices. Low-power microcontrollers ensure minimal energy consumption.
4. Transportation:
The project has potential applications in forested roadways or wildlife corridors, where it can alert drivers to the presence of animals near highways, reducing roadkill incidents and improving transportation safety.
5. Quality Education & Effective Governance:
The system can serve as a teaching tool for environmental education, promoting awareness about wildlife conservation and sustainable practices. It also aids policymakers in monitoring wildlife activity for better decision-making.
6. Communication & Networking (Cybersecurity):
With secure communication channels for transmitting video data and alerts, the system ensures data integrity and protection against unauthorized access.
7. Automation:
The project's use of sensors, AI models, and automated deterrent systems exemplifies a fully automated wildlife detection solution. It minimizes the need for manual monitoring, making it scalable and reliable for deployment in diverse environments.

This innovative solution contributes to environmental protection while improving agricultural and societal safety through intelligent automation and communication systems.

5. EXPECTED CONCLUSION

The WildEcoWatch system successfully integrates IoT and AI technologies to create a smart, efficient, and automated wildlife monitoring solution. Through the combined use of a PIR motion sensor, webcam, Raspberry Pi, and a custom-trained YOLOv8 model via Roboflow, the system can effectively detect and classify animals in real-time based on motion-triggered image capture.

Upon detection, the system is capable of providing immediate alerts through LED and buzzer activation, and more importantly, via automated email notifications to the concerned personnel using the Mailjet API. Furthermore, each detection event is logged into a structured CSV file, allowing long-term tracking, pattern analysis, and data-driven decisions for wildlife conservation efforts.

This project proves to be:

- Cost-effective, using affordable hardware.
- Scalable, for deployment in multiple forest zones or protected areas.
- Accurate, by filtering detections using AI with confidence thresholds.
- User-friendly, through a simple web interface for monitoring and manual prediction.

Ultimately, WildEcoWatch lays a strong foundation for future enhancements such as cloud-based data dashboards, solar-powered deployment, night vision support, and integration with remote IoT gateways. It represents a significant step forward in leveraging modern technology for the preservation of biodiversity and real-time wildlife surveillance.

6. REFERENCES:

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